

DENZEL CHINGODZA

Software Engineering Student · Full-Stack & AI / ML Development · Cape Town, South Africa

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· denz-platform.vercel.app

PROFESSIONAL SUMMARY

Final year BSc Information Technology (Software Engineering) student at Eduvos, Cape Town. Every tool in my stack came from building something real: I have independently shipped projects spanning AI/ML pipelines, serverless cloud infrastructure, RAG systems, and full-stack web applications. Seeking an internship or junior role in software engineering or AI/ML engineering where I can contribute to a real team and grow through doing actual work.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · SQL · HTML · CSS

Frameworks: FastAPI · Flask · Next.js · React · Tailwind CSS · SQLAlchemy

Databases: PostgreSQL · pgvector · MongoDB · SQLite · DynamoDB · Supabase · Neon

AI / ML: RAG pipelines · Vector embeddings · LLMs · Prompt engineering · PyTorch · HuggingFace Transformers · PEFT · QLoRA · NLP · OpenAI API

DevOps / Tools: Docker · Git · GitHub · GitHub Actions · REST APIs · AWS (Lambda, EventBridge, DynamoDB, SES, Cognito, API Gateway) · Terraform · Vercel · Render · Linux

EXPERIENCE

Full Stack Developer (Freelance)

2024 to Present

PBS Projects · Cape Town, South Africa

pbs-projects.vercel.app

- Designed and shipped a production web application for a real client, replacing a WhatsApp-based advertising operation with a full structured digital platform.
- Built a quote request pipeline with admin management, testimonial moderation, a product catalogue, and bilingual English/Shona support.
- Handled the entire engagement independently: requirements, design, implementation, and deployment to production.

Stack: Next.js · FastAPI · PostgreSQL · Cloudinary · Python

PROJECTS

Sentinel: Uptime Monitoring Platform

2025

sentinel-kappa-wine.vercel.app · github.com/denzelchingodza/Sentinel

- Monitors registered URLs every 60 seconds via scheduled AWS Lambda: on failure, an incident opens and an SES alert fires; on recovery, the incident closes and a recovery email follows, with full check history and response times stored in DynamoDB.
- Serverless backend on AWS: EventBridge schedules the monitor Lambda, API Gateway with Cognito JWT auth fronts the API Lambda, and three DynamoDB tables store monitors, check history, and incidents.
- Provisioned all infrastructure with Terraform (Lambda, EventBridge, API Gateway, DynamoDB, SES, Cognito), all reproducible from a single deploy. First time working with IaC; changed how I think about infrastructure permanently.
- Next.js 15 (App Router, TypeScript) frontend with full auth lifecycle (sign-up, email verify, sign-in, password reset, account deletion) deployed on Vercel.

Stack: Node.js · AWS Lambda · EventBridge · DynamoDB · SES · Cognito · API Gateway · Terraform · Next.js · TypeScript · Vercel

Muzezuru: Shona Language AI

2025 to Present

muzezuru.vercel.app · github.com/denzelchingodza/muzezuru

- Fine-tuned BLOOMZ-3B on Shona, a severely underrepresented low-resource African language, using QLoRA with 4-bit quantization entirely on free-tier compute (Google Colab, Kaggle, HuggingFace ZeroGPU).
- Discovered and resolved an 84% data contamination issue in the primary training dataset before retraining on the cleaned corpus, a find that materially changed the model's output quality.
- Built a custom 15-prompt evaluation scorecard across 7 categories, since no standard Shona NLP benchmark exists, then shipped a full Next.js chat frontend with streaming responses from the HuggingFace Gradio API.

Stack: Python · PyTorch · BLOOMZ-3B · QLoRA · PEFT · bitsandbytes · HuggingFace Transformers · NLP · Next.js · TypeScript

DocuZen: AI Document Q&A

2025

doc-analyzer-as5k.vercel.app · github.com/denzelchingodza/doc-analyzer

- Full RAG pipeline from scratch: PDF/DOCX ingestion, token-aware chunking with tiktoken, text-embedding-3-small for vectors, cosine-similarity retrieval via pgvector, and GPT-4o-mini for grounded answers with page-level citations.
- FastAPI backend with async SQLAlchemy, rate limiting (60 req/hr per IP), CORS restrictions, and 50 MB file validation; served via Docker on Render against a Neon serverless PostgreSQL database with pgvector.
- Next.js 15 frontend with a 3-panel interface (document sidebar, PDF viewer, chat panel) deployed on Vercel.

Stack: Python · FastAPI · PostgreSQL · pgvector · OpenAI API · tiktoken · Next.js · TypeScript · Docker · Render · Vercel

StackScope: Job Market Intelligence

2025

stack-scope-eight.vercel.app · github.com/denzelchingodza/StackScope

- Data pipeline scraping live developer job postings from 5 boards (Adzuna, Remoteive, We Work Remotely, Jobspresso, JobLeads), storing structured records in Supabase PostgreSQL.
- Keyword scoring engine using TF-IDF and cosine similarity ranks listings against a user's skill set and surfaces a gap analysis showing which skills appear most in postings but are absent from their stack.
- Per-currency salary stats (USD, ZAR, GBP, EUR) with automatic detection from raw salary strings and breakdowns by experience level.

Stack: Python · Flask · BeautifulSoup · Supabase · PostgreSQL · JavaScript · Vercel · Render

WHAT I BRING

- **I build to learn.** Every tool in my stack was picked up through a real project, not a tutorial. Self-directed learner.
- **I ship.** All projects are live, publicly accessible, and maintained. I take things to completion.
- **Strong fundamentals.** Built REST APIs, relational databases, async Python, full-stack apps, and ML pipelines without scaffolding tools.
- **I want to grow fast.** I am actively looking for a team where I can learn from engineers who are further along than me.

EDUCATION

BSc Information Technology, Software Engineering

2022 to Present

Eduvos · Cape Town, South Africa · Expected graduation: 2026 · Final year

- Relevant coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Operating Systems, Computer Networks, Object-Oriented Programming.
- Consistently applied academic concepts in self-initiated projects beyond coursework, treating every module as a foundation for something buildable.

References available on request · Portfolio and all live projects at denz-platform.vercel.app